

A Mini Project Report on

TABLEAU: FROM VISUALIZATION TO DASHBOARD PUBLISHING & ANALYSIS

**A Study of Business Intelligence Visualization, Dashboard Design and
Publishing Using the Sample Superstore Dataset**

**Submitted in partial fulfilment of the requirements for the award of the
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Business Analytics

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CERTIFICATE

This is to certify that the mini project report titled “Tableau: From Visualization to Dashboard Publishing & Analysis” has been carried out and submitted by Rakshitha M M, in partial fulfilment of the requirements for the award of the degree of Master of Business Administration (Business Analytics), Nagarjuna Degree College, under my guidance and supervision.

The report is a genuine record of the practical work, secondary research and dashboard analysis carried out by the candidate for the Business Analytics course, and to the best of my knowledge has not been submitted elsewhere for the award of any other degree or diploma.

This project is forwarded for evaluation.

Place: Bengaluru

Date:

Signature of Guide

Assistant Professor Archana Karnik K.M

DECLARATION

I, Rakshitha M M, hereby declare that the mini project report titled “Tableau: From Visualization to Dashboard Publishing & Analysis” is an original record of work carried out by me under the guidance of Assistant Professor Archana Karnik K.M, Department of Management, Nagarjuna Degree College, in partial fulfilment of the Business Analytics course requirements of the MBA programme.

The dataset used for the practical exercises in this report is the publicly available Sample Superstore dataset, used purely for academic learning and demonstration of Tableau features. All dashboard implementations referenced in this report are drawn from my own project repositories.

I further declare that this work has not been submitted previously, in full or in part, for the award of any degree, diploma, associateship or other similar title of this or any other university or examining body.

Place: Bengaluru

Date:

Rakshitha M M

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I also thank my classmates and friends for their useful discussions on Tableau features and dashboard design practices, which helped sharpen the analysis presented in this report.

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EXECUTIVE SUMMARY

This mini project studies Tableau as a business intelligence and data-visualization platform, moving from the fundamentals of chart construction to the design, publishing and analysis of complete interactive dashboards. Using the widely adopted Sample Superstore dataset, the project builds eighteen individual worksheets covering text tables, heat maps, highlight tables, geographic maps, bar and line charts, dual-axis combinations, pie charts, scatter plots, histograms, Gantt charts, filters and Key Performance Indicator (KPI) cards, following a structured practical brief.

These eighteen worksheets are consolidated into two published dashboards — the first assembling sheets 1 to 9 and the second assembling sheets 10 to 18 — both published to Tableau Public, in line with the practical requirement of the course. The report also draws on two independent dashboard projects developed by the author, the Tableau Sales Intelligence Dashboard and the Product Sales Analysis Dashboard, to illustrate how the same visualization building blocks are assembled into executive-facing, decision-support tools.

The study finds that the value of Tableau lies not in any single chart type but in the way interactive filters, dashboard actions and KPI cards let a business user move from a high-level summary to a granular, root-cause view without writing a single query. The report closes with a discussion of Tableau's advantages and limitations relative to traditional static reporting, and with recommendations on how the two published dashboards can be extended for real managerial use.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
BI	Business Intelligence
KPI	Key Performance Indicator
VizQL	Visual Query Language (Tableau's underlying query engine)
CSV	Comma-Separated Values
TDE / Hyper	Tableau Data Extract / Tableau's in-memory data engine format
MBA	Master of Business Administration
ETL	Extract, Transform, Load
UI/UX	User Interface / User Experience
YoY	Year-over-Year

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CHAPTER 1 — INTRODUCTION TO TABLEAU

1.1 Introduction

Tableau is a leading Business Intelligence (BI) and data-visualization platform that allows users to connect to data of almost any size or shape, explore it visually, and turn it into interactive dashboards without writing SQL or code. It replaces the traditional, engineer-dependent reporting cycle — where a business user waits for an analyst to run a query and export a static chart — with a drag-and-drop interface that lets the business user ask a new question and see the answer in seconds.

At its core, Tableau translates simple visual actions (dragging a field onto a shelf, choosing a mark type, applying a filter) into an underlying query language, VizQL, which is executed against the connected data source and rendered back as a chart. This report studies Tableau from first principles — what it is, how it is structured, how charts and dashboards are built — and then applies that theory to a practical dataset and to two real dashboard projects.

1.2 Evolution of Data Visualization

Business reporting has moved through three broad stages. The first stage was the static spreadsheet and printed report: numbers were tabulated in Excel, formatted manually, and circulated as a fixed snapshot that was already outdated by the time it reached its reader. The second stage introduced templated dashboards built by IT teams using early reporting tools, which improved consistency but still required a change request and a development cycle for every new question.

The third and current stage is self-service, interactive Business Intelligence, of which Tableau is one of the most prominent examples. Here, the business user explores the data directly: filtering, drilling down, and re-arranging a view in real time. This shift places analytical power directly in the hands of the people making decisions, and is the central reason Tableau and similar tools have become standard in modern business analytics curricula.

1.3 What is Tableau?

Tableau is not a single application but a family of related products that together form an end-to-end analytics ecosystem:

- Tableau Desktop — the authoring application used to connect to data, build worksheets, dashboards and stories.
- Tableau Public — a free version of Tableau used to publish dashboards to a public online gallery; used for the practical work in this report.
- Tableau Server — an on-premises platform that lets an organisation publish, secure and share dashboards internally.
- Tableau Cloud — the fully-hosted, cloud-based equivalent of Tableau Server, removing the need for the organisation to manage its own infrastructure.
- Tableau Prep — a dedicated data-preparation tool used to clean, shape and combine data before it reaches Tableau Desktop.

- Tableau Reader — a free viewer that lets a recipient open a packaged Tableau workbook (.twbx) without owning a Desktop licence.

1.4 Importance of Tableau in Business Analytics

Business Analytics is concerned with turning raw operational data into decisions. Tableau supports every stage of that journey: it connects to the raw data, helps the analyst explore it visually to find patterns that would be invisible in a spreadsheet, and then packages those findings into a dashboard that a non-technical manager can use unaided. Because dashboards refresh automatically when the underlying data updates, decision-makers work from current information rather than a report that was accurate only on the day it was generated.

1.5 Applications of Tableau

Tableau is used across nearly every business function:

- Sales Analytics — tracking revenue, quota attainment and pipeline by region, product and salesperson.
- Marketing Analytics — campaign performance, channel attribution and customer acquisition cost.
- Financial Analytics — budget-vs-actual tracking, profitability analysis and financial ratio dashboards.
- HR Analytics — headcount, attrition, diversity and workforce-planning dashboards.
- Operations — production throughput, quality metrics and process bottleneck analysis.
- Customer Analytics — segmentation, lifetime value and churn-risk visualization.
- Supply Chain — inventory levels, supplier performance and logistics tracking.
- Management Reporting — consolidated executive dashboards spanning multiple functions.

1.6 Tableau in Decision Making

Visualization improves decision-making because the human visual system processes a well-designed chart far faster than it processes a table of numbers. A manager scanning a Tableau dashboard can spot an underperforming region through colour and position within seconds, then click that region to see exactly which product or customer segment is driving the shortfall. This combination of speed (fast pattern recognition) and depth (interactive drill-down) is what distinguishes Tableau-based decision-making from decision-making based on static monthly reports.

CHAPTER 2 — OBJECTIVES OF TABLEAU

The broad objectives that a business pursues in adopting Tableau, and that this project pursues in studying it, are set out below.

1. Transform raw, row-level data into meaningful, interpretable visualizations.
2. Identify trends and patterns that are not visible in tabular data.
3. Compare business performance across regions, products, segments and time periods.
4. Analyse sales and profitability drivers at a granular level.
5. Create interactive dashboards that support self-service exploration.
6. Support data-driven, rather than intuition-driven, decision-making.
7. Enable geographical analysis of business performance using maps.
8. Create Key Performance Indicators (KPIs) that summarise business health at a glance.
9. Publish dashboards so that they can be shared and consumed by stakeholders.
10. Communicate business insights effectively to both technical and non-technical audiences.

2.1 Primary Objectives

- To understand Tableau's architecture, components and workbook structure.
- To build and interpret the principal chart types used in business analytics.
- To design, publish and analyse two complete interactive dashboards.

2.2 Secondary Objectives

- To evaluate the advantages and limitations of Tableau relative to traditional reporting.
- To document a reusable practical reference for future Tableau coursework.

2.3 Business Objectives

- To convert the Sample Superstore transactional data into an executive-ready view of sales, profit and regional performance.
- To surface actionable business insights — such as loss-making segments or high-performing regions — that a manager could act on directly.

2.4 Learning Objectives of the Project

- To gain hands-on proficiency with Tableau Desktop's core chart-building workflow.
- To learn dashboard design principles, including layout, interactivity and filter actions.
- To learn the end-to-end publishing workflow to Tableau Public.

CHAPTER 3 — TABLEAU ARCHITECTURE

3.1 Overview of Tableau Architecture

Tableau's architecture is organised around a clear separation between where data lives, where visualizations are authored, and where they are consumed. A workbook built in Tableau Desktop can connect live to a database or work from an in-memory extract, and once built, that workbook can be published outward to Tableau Server, Tableau Cloud, or Tableau Public depending on the intended audience.

3.2 Tableau Components

- Tableau Desktop — the authoring layer where worksheets, dashboards and stories are built.
- Tableau Server — the internal publishing and governance layer for an organisation.
- Tableau Cloud — the SaaS equivalent of Tableau Server, hosted by Salesforce/Tableau.
- Tableau Public — the free, public publishing destination used for this project's dashboards.
- Tableau Prep — the upstream data-preparation layer.
- Tableau Bridge — a connector that lets cloud-hosted Tableau reach data sitting behind an organisation's firewall.

3.3 Tableau Data Flow

The path a dataset takes from its raw source to a business user's screen follows a consistent sequence:

Figure 3.1 — Tableau Data Flow: Data Sources → Data Connection → Data Preparation → Tableau Desktop → Worksheets → Dashboards → Stories → Publishing → Tableau Server / Cloud / Public → Business Users

Each stage adds structure: the raw source is connected to, cleaned and shaped in preparation, assembled into individual worksheets, combined into dashboards, optionally sequenced into a narrative story, and finally published to the platform appropriate to the audience — internal (Server/Cloud) or public (Tableau Public, as used in this project).

3.4 Live Connection vs Extract

Tableau offers two fundamentally different ways of connecting to data, and the choice between them affects performance, freshness and offline availability.

Aspect	Live Connection	Extract (Hyper)
Data freshness	Always current — queries the source directly	As current as the last extract refresh
Performance	Depends on source database speed	Generally faster — optimised in-memory engine
Offline use	Not possible without the source	Workbook can be opened without the source
Best suited for	Frequently changing, large enterprise databases	Moderate-sized datasets like Sample Superstore

Table 3.1 — Live Connection vs. Extract

For the practical work in this report, an extract-based connection to the Sample Superstore dataset was used, since the dataset is static and an extract gives faster interaction while building the eighteen worksheets.

3.5 Tableau Workbook Structure

- Data Sources — one or more connections feeding the workbook, visible as tabs at the bottom of the screen.
- Worksheets — individual charts, each built from fields dragged onto Rows, Columns, Marks and Filters.
- Dashboards — a canvas combining multiple worksheets, filters and objects into one interactive view.
- Stories — a sequence of dashboard/worksheet snapshots used to narrate an analysis step by step.

CHAPTER 4 — DATA PREPARATION

4.1 Dataset Introduction

All practical work in this report uses the Sample Superstore dataset, a widely used retail transactional dataset bundled with Tableau and adopted throughout business-analytics coursework worldwide. It records individual order-line transactions for a fictional retail superstore across the United States, spanning several years of sales, profit and shipping data.

4.2 Data Source

Sample Superstore ships as a native Tableau data source (and an equivalent Excel workbook) and was connected to directly within Tableau Desktop for this project, using an extract connection as discussed in Chapter 3.

4.3 Dataset Structure

The dataset is organised as one row per order line, with the following key fields used throughout the practical experiments:

Field	Type	Description
Order ID	Dimension	Unique identifier for each customer order
Order Date	Dimension (Date)	Date the order was placed
Ship Date	Dimension (Date)	Date the order was shipped
Customer	Dimension	Name of the purchasing customer
Segment	Dimension	Consumer, Corporate or Home Office
Region	Dimension	Sales region (East, West, Central, South)
State	Dimension	US state of the customer
Category	Dimension	Furniture, Office Supplies or Technology
Sub-Category	Dimension	Finer product grouping within Category
Sales	Measure	Revenue generated by the order line
Quantity	Measure	Units sold in the order line
Discount	Measure	Discount rate applied to the order line
Profit	Measure	Profit (or loss) generated by the order line

Table 4.1 — Key Fields of the Sample Superstore Dataset

4.4 Data Cleaning

Sample Superstore is a well-curated teaching dataset, but the same checks applied to any real dataset were run before building worksheets:

- Missing values — the Sales, Profit, Quantity and Discount measures were checked for nulls; none were found in the standard extract.

- Duplicate records — Order ID together with the product line was checked for unintended duplication.
- Incorrect data types — Order Date and Ship Date were confirmed as Date fields rather than text, so that Tableau's automatic date-hierarchy features (Year, Quarter, Month) would be available.
- Date formatting — both date fields were confirmed to use a consistent format on import.
- Numeric fields — Sales, Profit and Discount were confirmed as continuous numeric measures rather than strings.
- Categorical fields — Region, Category, Sub-Category and Segment were confirmed as discrete dimensions.

4.5 Data Preparation Process

The preparation workflow followed for this project was: connect to the Sample Superstore data source, inspect the automatically assigned data types, verify the dimension/measure split described in Section 4.6, and take an extract before beginning worksheet design, so that the eighteen practical experiments in Chapter 9 would run against a fast, locally cached copy of the data.

4.6 Dimensions and Measures

Tableau automatically classifies every field as either a dimension (a categorical field used to slice and group data, such as Region or Category) or a measure (a numeric field that is aggregated, such as Sales or Profit). This classification determines default behaviour on the Rows and Columns shelves and is central to every worksheet built in Chapter 9 of the practicals.

4.7 Discrete and Continuous Fields

Independently of the dimension/measure split, every field is also either discrete (shown as a distinct, unordered header — coloured blue in Tableau's field list) or continuous (shown along a measurable axis — coloured green). Order Date, for example, can be used either discretely (as a header, e.g. Year(Order Date)) or continuously (as a trend axis), and both usages appear across the eighteen practical experiments.

4.8 Filters and Parameters

Filters restrict the data shown in a view — for example, limiting a worksheet to a single Region — and are used in Experiment 17 of the practicals. Parameters are user-defined, dynamic values (such as a selectable top-N cut-off) that can drive calculated fields, titles and filters; they extend interactivity beyond what a simple filter can do, letting a single worksheet answer a family of related questions.

4.9 Data Quality Considerations

Because every downstream chart and KPI depends on the underlying data being accurate and consistently typed, the checks in Section 4.4 were treated as a mandatory first step rather than an optional one. In a live business setting, this stage would also include reconciliation against a source system of record and a documented data dictionary — both of which sit outside the scope of a teaching dataset such as Sample Superstore, but are noted here for completeness.

CHAPTER 5 — CHARTS AND GRAPHS IN TABLEAU

This chapter explains, chart by chart, the visualization types built as part of the eighteen practical experiments documented in full in the Practical Programs section of this report. For each chart type the definition, the field-placement procedure (as followed in the practical work), its typical business use, and the kind of insight it yields are set out below. The exact field-placement steps used in the hands-on experiments are reproduced faithfully from the practical brief and are not altered here.

1. Text Table

A Text Table is the simplest Tableau view: a grid of raw or aggregated numbers, functionally similar to a pivot table.

Procedure:

- Category → Rows
- Sales → Text
- Select Show Me → Text Table

Business Use: Quick, precise numeric lookups where exact values matter more than visual pattern.

Insight: Compares category-wise sales at a glance.

2. Heat Map

A Heat Map uses colour intensity across a grid of two dimensions to show the magnitude of a measure.

Procedure:

- Region → Columns
- Category → Rows
- Sales → Color
- Select Heat Map

Business Use: Spotting which region/category combinations dominate sales without reading numbers.

Insight: Identifies high-performing regions.

3. Highlight Table

A Highlight Table is a Text Table with cell-level colour shading layered on top, combining exact figures with visual scanning.

Procedure:

- Region → Columns
- Category → Rows
- Profit → Color
- Select Highlight Table

Business Use: Finding profitable versus loss-making cells while still reading the underlying number.

Insight: Locates profitable and loss-making segments.

4. Symbol Map

A Symbol Map plots geocoded points (such as states) on a map, sizing and colouring each symbol by a measure.

Procedure:

- State → Detail
- Sales → Size
- Profit → Color
- Show Me → Symbol Map

Business Use: Understanding geographic concentration of sales and profitability simultaneously.

Insight: Reveals geographical sales distribution.

5. Bar Chart

A Bar Chart compares a measure across the members of a dimension using bar length.

Procedure:

- Category → Rows
- Sales → Columns
- Show Me → Bar Chart

Business Use: The default choice for ranking categories by performance.

Insight: Compares category performance.

6. Filled Map

A Filled Map shades entire geographic regions (e.g. whole states) by a measure, rather than plotting a point symbol.

Procedure:

- State → Detail
- Sales → Color
- Show Me → Filled Map

Business Use: Communicating broad geographic patterns to a non-technical audience.

Insight: Shows state-wise market penetration.

7. Stacked Bar Chart

A Stacked Bar Chart shows a total bar broken into coloured segments representing a second dimension.

Procedure:

- Category → Rows

- Sales → Columns
- Segment → Color
- Show Me → Stacked Bars

Business Use: Seeing both the total and its composition in one view.

Insight: Shows segment contribution to sales.

8. Side-by-Side Bar Chart

A Side-by-Side Bar Chart places bars for a second dimension next to each other rather than stacked, making direct comparison easier.

Procedure:

- Category → Rows
- Sales → Columns
- Segment → Columns
- Show Me → Side-by-Side Bars

Business Use: Comparing segment performance within each category precisely.

Insight: Enables segment comparison.

9. Line Chart

A Line Chart plots a continuous measure against a continuous time axis to show trend over time.

Procedure:

- Order Date → Columns
- Sales → Rows
- Show Me → Line Chart

Business Use: The standard chart for tracking performance trends month over month or year over year.

Insight: Supports sales trend analysis.

10. Dual Axis / Dual Line Chart

A Dual Axis (Dual Line) Chart overlays two measures with independent axes on the same set of category headers, so their trends can be compared directly.

Procedure:

- Order Date → Columns
- Sales → Rows
- Profit → Rows
- Combine as Dual Axis

Business Use: Checking whether rising sales are actually translating into rising profit.

Insight: Shows the relationship between sales and profit.

12. Pie Chart

A Pie Chart shows each category's share of a whole using the angle of a wedge.

Procedure:

- Category → Color
- Sales → Angle
- Show Me → Pie Chart

Business Use: Communicating simple part-to-whole composition to a non-technical audience.

Insight: Shows category contribution to total sales.

13. Scatter Plot

A Scatter Plot places two continuous measures on perpendicular axes, with each mark representing one member of a dimension, to reveal correlation and outliers.

Procedure:

- Sales → Columns
- Profit → Rows
- Category → Color
- Show Me → Scatter Plot

Business Use: Spotting categories that generate high sales but disproportionately low (or negative) profit.

Insight: Supports profitability analysis.

14. Circle View

A Circle View is a compact chart that encodes one measure through the size of a circle and a second through its colour, arranged against a dimension.

Procedure:

- Category → Rows
- Sales → Size
- Profit → Color
- Show Me → Circle View

Business Use: Getting a compact, two-measure comparison across many categories.

Insight: Provides a visual comparison of two measures at once.

15. Histogram

A Histogram groups a continuous measure into equal-width bins and shows the count of records falling into each bin, revealing the shape of a distribution.

Procedure:

- Profit → Select the field

- Show Me → Histogram

Business Use: Understanding whether profit is concentrated near zero, widely spread, or skewed by a few large losses.

Insight: Shows the distribution of profit across order lines.

16. Gantt Chart

A Gantt Chart uses horizontal bars positioned along a time axis to show duration or sequence, most often associated with project or order timelines.

Procedure:

- Order ID → Rows
- Sales → Size
- Show Me → Gantt Chart

Business Use: Comparing orders by size and position along a timeline in a single view.

Insight: Supports order-level comparison.

17. Filters

Filters restrict a view to a chosen subset of the data — either applied to a single worksheet or, when placed on a dashboard, made available for interactive use by a viewer.

Procedure:

- Drag Region to Filters.
- Select the regions to include.
- Show Filter (so the control appears on screen).

Outcome: Enables interactive dashboard filtering, letting a viewer narrow every worksheet on a dashboard to a single region with one click.

18. KPI Cards — Creating Business KPIs in Tableau

A KPI Card is a large, single-number worksheet used to headline a critical business metric — in this project, Total Sales — on a dashboard.

Procedure (KPI: Total Sales):

- Open a new worksheet and rename it “KPI - Total Sales”.
- Drag Sales from Measures to the Text mark card.
- Tableau displays the aggregated total sales value.
- Click the Sales field on the Marks card → Format.
- Format the number as Currency (Custom).
- Increase the font size (20–30 pt) and make it bold.

Business Use: Headline metrics for executive dashboards, giving a viewer the single most important number before they explore further detail.

CHAPTER 6 — DASHBOARDS AND STORIES

6.1 What is a Tableau Dashboard?

A Tableau Dashboard is a single canvas that combines multiple worksheets, filters, legends and other objects into one interactive view. Rather than viewing eighteen separate worksheets one at a time, a dashboard presents the relevant subset together, arranged so that a viewer can see the whole picture and drill into any part of it without leaving the screen.

6.2 Dashboard Components

- Worksheets — the individual charts placed on the dashboard canvas.
- Filters — interactive controls that narrow the data shown across one or more worksheets.
- Legends — colour/size keys that explain how a measure or dimension is encoded.
- Text and Title objects — headings, notes and instructions for the viewer.
- Images and Web Page objects — logos, icons or embedded external content.
- Layout containers — horizontal and vertical containers that control how objects resize together.

6.3 Dashboard Design Principles

- Lead with the headline number (a KPI card) before supporting detail.
- Group related worksheets together and keep a consistent colour encoding across them.
- Limit each dashboard to a focused set of questions rather than crowding in every possible chart.
- Use white space deliberately so the eye is not overwhelmed.
- Design for the target device — a fixed desktop layout differs from a fluid, mobile-first one.

6.4 Interactive Filters

Filters placed on a dashboard (Experiment 17 of the practicals) can be applied to a single worksheet or, using “Apply to Worksheets → All Using This Data Source”, to every worksheet on the dashboard simultaneously, so that selecting a single Region instantly updates every chart in view.

6.5 Dashboard Actions

Dashboard Actions go one step further than filters: clicking a mark in one worksheet (a Filter Action), hovering over it (a Highlight Action), or selecting it (a Go-to-URL or Go-to-Sheet Action) can drive a response elsewhere on the dashboard. This is what makes a Tableau dashboard feel like a single connected tool rather than a collection of separate charts placed next to each other.

6.6 KPI Cards

As built in Experiment 18 of the practicals, a KPI Card surfaces one headline number — here, Total Sales — in large, bold, currency-formatted text, giving the dashboard viewer an immediate anchor figure before they explore the supporting charts.

6.7 Creating an Executive Dashboard

An executive dashboard is typically assembled by placing the KPI card at the top, a trend chart (Experiment 9's Line Chart) beneath it, a ranked comparison (Experiment 5's Bar Chart or Experiment 1's Text Table) to one side, and a geographic view (Experiment 4 or 6) to ground the numbers in a map, all tied together with a Region filter (Experiment 17) so a single control updates the whole view.

6.8 Tableau Stories

A Tableau Story is a sequence of dashboard or worksheet “snapshots”, each captioned, that lets an analyst walk an audience through an analysis step by step — for example, starting from an overall sales trend, then narrowing into the region driving a dip, then into the specific sub-category responsible.

6.9 Dashboard vs Story

Aspect	Dashboard	Story
Purpose	Ongoing, self-service exploration	Guided, sequential narrative
Interactivity	Fully interactive — filters, actions	Interactive within each snapshot, but moves in a set sequence
Best for	Recurring monitoring by a business user	Presenting findings to a specific audience
Typical audience	Analysts, managers who self-serve	Executives, stakeholders in a meeting

Table 6.1 — Dashboard vs. Story

PROJECT DASHBOARDS — PRACTICAL IMPLEMENTATION REFERENCE

Two independent dashboard projects, developed by the author and hosted on project repository, are used as the practical dashboard-implementation reference for this report. Both apply the chart types documented in Chapter 5 to the Sample Superstore dataset, consistent with the practical brief that groups Experiments 1–9 into a first dashboard and Experiments 10–18 into a second.

Dashboard 1 — Tableau Sales Intelligence Dashboard

Repository link: to be added by the student.

This dashboard is positioned, by its name, as a sales-intelligence tool — the natural home for the KPI card (Experiment 18), the sales trend line (Experiment 9), the category/region comparisons (Experiments 1, 2, 5, 7, 8) and the interactive Region filter (Experiment 17) documented in this report. At the time of writing, the repository link was not supplied for independent verification while preparing this report, so its exact worksheet layout, specific KPIs and filter configuration are not independently verified here.

To keep this report's claims fully aligned with the academic rule against inventing unverified features, the detailed breakdown below is left for the student to complete directly from the live Tableau Public workbook and project README before submission:

- Dashboard purpose — to be inserted by student, drawn from the repository README.
- KPIs shown — to be inserted by student.
- Visualizations used — to be inserted by student, matched against the chart types in Chapter 5.
- Filters available — to be inserted by student.
- Business questions answered — to be inserted by student.
- Business insights and decision-making value — to be inserted by student.

Dashboard 2 — Product Sales Analysis Dashboard

Repository link: to be added by the student.

This dashboard is positioned, by its name, as a product-level analysis tool — a natural home for the profitability views (Experiments 3, 13, 15), the geographic views (Experiments 4, 6) and the composition charts (Experiments 7, 8, 12) documented in this report. As with Dashboard 1, the live repository content could not be independently verified through the tools available while preparing this report, so the specific breakdown below should be completed by the student from the actual published workbook:

- Dashboard purpose — to be inserted by student.
- KPIs shown — to be inserted by student.
- Visualizations used — to be inserted by student.
- Filters available — to be inserted by student.
- Business questions answered — to be inserted by student.
- Business insights and decision-making value — to be inserted by student.

Note on academic accuracy: per the project brief, this report does not claim that either dashboard contains a specific feature unless that feature is directly confirmed from the source project. The structural analysis above is therefore expressed in terms of what each dashboard is positioned to contain, based on its title and on the practical chart set in Chapter 5, with the verified specifics left as an explicit action item for the student.

CHAPTER 7 — ADVANTAGES AND DISADVANTAGES OF TABLEAU

7.1 Advantages

- Ease of visualization — charts are built by dragging fields onto shelves rather than writing code.
- Interactive dashboards — filters and actions let a viewer explore data themselves.
- Drag-and-drop interface — lowers the technical barrier for business users.
- Large data-source support — connects to spreadsheets, databases, cloud warehouses and web data.
- Advanced analytics — trend lines, forecasting and statistical summaries are built in.
- Geographic visualization — native mapping without a separate GIS tool.
- Dashboard sharing — publishing to Server, Cloud or Public makes distribution simple.
- Tableau Public — a free route to publish and showcase work, as used in this project.
- Business-user accessibility — designed for analysts and managers, not only engineers.
- Fast exploratory analysis — a new question can be answered in seconds, not a development cycle.

7.2 Disadvantages

- Licensing cost — Tableau Desktop, Server and Cloud carry a significant per-user licence cost for a full organisation.
- Learning curve for advanced features — calculated fields, level-of-detail (LOD) expressions and dashboard actions take time to master.
- Large workbooks can become complex — many worksheets and dashboards in one workbook can be hard to navigate.
- Data preparation may require additional tools — heavy cleaning is often better done in Tableau Prep or upstream of Tableau.
- Tableau Public privacy limitations — workbooks published to Tableau Public, and their underlying data, are visible to anyone.
- Advanced calculations require technical knowledge — complex LOD expressions and table calculations have a real learning curve.

7.3 Tableau Compared with Traditional Reporting

Aspect	Traditional Reporting (Spreadsheets/Static Reports)	Tableau
Update cycle	Manual refresh, often monthly	Live or scheduled automatic refresh
Interactivity	None — fixed snapshot	Fully interactive filters and drill-down
Turnaround for new questions	Days to weeks (IT/analyst request)	Minutes (self-service)

Aspect	Traditional Reporting (Spreadsheets/Static Reports)	Tableau
Visual quality	Basic charts, inconsistent formatting	Purpose-built, consistent visual design
Scalability	Breaks down with large data volumes	Built for large and growing datasets
Audience reach	Emailed file, single point-in-time view	Published, shareable, always-current dashboard

Table 7.1 — Tableau Compared with Traditional Reporting

CHAPTER 8 — CONCLUSION

This project set out to move from the fundamentals of Tableau as a visualization platform to the practical construction and publishing of complete interactive dashboards. Across eighteen guided practical experiments, every major Tableau chart type — text tables, heat maps, highlight tables, geographic maps, bar, line and pie charts, dual-axis combinations, scatter plots, histograms, Gantt charts, filters and KPI cards — was built against the Sample Superstore dataset and consolidated into two published dashboards.

The exercise reinforced a central lesson of business analytics: a chart is only useful insofar as it answers a specific business question. Sales Analytics, Marketing Analytics, Financial Analytics and the other application areas surveyed in Chapter 1 all rest on the same underlying skill — choosing the right visual encoding for the question being asked — which this project practised systematically, chart type by chart type.

Building and interpreting the eighteen experiments also demonstrated Tableau's role as a decision-support tool rather than merely a charting tool: the KPI card, dashboard filters and dashboard actions built in Chapter 6 turn a set of individual charts into a single, connected instrument that a manager can operate without further training. The two independently hosted dashboard projects referenced in this report — the Sales Intelligence Dashboard and the Product Sales Analysis Dashboard — illustrate how the same building blocks scale up into dedicated, purpose-built tools for a specific business audience.

On the practical skills side, the project built genuine, transferable competence: connecting to a data source and choosing between a live connection and an extract, classifying and using dimensions, measures, discrete and continuous fields correctly, constructing each of the eighteen chart types from the correct field placement, assembling worksheets into an interactive dashboard, and publishing that dashboard to Tableau Public for public access.

8.1 Findings

- Interactive filters and dashboard actions materially increase the analytical value of a set of charts compared with the same charts viewed individually.
- Geographic charts (Symbol Map, Filled Map) add a dimension of insight — spatial concentration — that tabular views cannot convey.
- A single, well-placed KPI card gives a dashboard an immediate point of reference that supporting charts can then be read against.
- Grouping Experiments 1–9 and 10–18 into two separate, focused dashboards (as specified in the practical brief) produces a cleaner, less crowded result than a single dashboard carrying all eighteen worksheets.

8.2 Recommendations

- Extend both published dashboards with a documented README on project repository describing their exact KPIs, filters and intended audience, so the analysis in Chapter 6 of this report can be verified and cited precisely.
- Add dashboard actions (filter and highlight actions) between the worksheets on each dashboard, if not already present, to increase interactivity beyond a single Region filter.

- Where the dataset allows, add a forecasting or trend-line layer to the sales trend chart to give the dashboard a forward-looking, not only historical, view.

8.3 Future Scope

Future extensions of this project could connect Tableau to a live, refreshing data source rather than a static extract, incorporate Tableau Prep for more advanced data-blending scenarios, and explore Tableau's integration with predictive-analytics workflows (for example, forecasting or clustering) to move the dashboards from a purely descriptive to a more predictive orientation. Publishing to Tableau Server or Tableau Cloud, rather than only Tableau Public, would also be a natural next step for a workplace setting where the underlying data cannot be made publicly visible.

TABLEAU PRACTICAL PROGRAMS / EXPERIMENTS

Aim of the Practical Set: To design two dashboards using the Sample Superstore dataset — including Sheets 1 to 9 in the first dashboard and Sheets 10 to 18 in the second dashboard — and to publish both dashboards to Tableau Public and share them, following the practical brief provided for this course.

Dataset used: Sample Superstore.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
1	US-2023-101	05-01-2023	07-01-2023	Standard Class	DP-1000	Daren Powers	Consumer	United States	Houston	Texas	77095	Central	OFF-PA-1000	Office Supply	Paper	Message Book-Vlr	16.448	2	0.2	5.5512
2	US-2023-111	04-01-2023	08-01-2023	Standard Class	PO-1995	Phyllis Ober	Home Office	United States	Naperville	Illinois	60540	Central	OFF-BI-1000	Office Supply	Binders	GBG Standard Plan	3.54	2	0.8	-5.487
3	US-2023-111	04-01-2023	08-01-2023	Standard Class	PO-1995	Phyllis Ober	Home Office	United States	Naperville	Illinois	60540	Central	OFF-LA-1000	Office Supply	Labels	Avery 500	31794	3	0.2	4.2707
4	US-2023-111	04-01-2023	08-01-2023	Standard Class	PO-1995	Phyllis Ober	Home Office	United States	Naperville	Illinois	60540	Central	OFF-ST-1000	Office Supply	Storage	SAFECO Ballers St	272.736	3	0.2	-64.775
5	US-2023-141	05-01-2023	11-01-2023	Standard Class	ME-1005	Mick Brown	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-AR-1000	Office Supply	Art	Avery H-Line Everl	19.536	3	0.2	4.884
6	US-2023-141	06-01-2023	07-01-2023	Standard Class	ME-17320	Maria Eneadi	Home Office	United States	Henderson	Kentucky	42420	South	FUR-CH-1000	Furniture	Chairs	Global Deluxe High	2573.62	9	0	746.450
7	US-2023-141	06-01-2023	07-01-2023	Standard Class	ME-17320	Maria Eneadi	Home Office	United States	Henderson	Kentucky	42420	South	OFF-AR-1000	Office Supply	Art	Rogers Handheld E	5.48	2	0	14.786
8	US-2023-104	06-01-2023	07-01-2023	First Class	JO-6145	Jack O'Brian	Corporate	United States	Atlanta	Georgia	30605	South	OFF-AR-1000	Office Supply	Art	Dixon Prang Vases	12.78	3	0	5.2336
9	US-2023-141	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Eneadi	Home Office	United States	Henderson	Kentucky	42420	South	OFF-BI-1000	Office Supply	Binders	Bacio Hk Tech Mats	603.38	2	0	274.491
10	US-2023-141	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Eneadi	Home Office	United States	Henderson	Kentucky	42420	South	OFF-FA-1000	Office Supply	Furniture	Alliance Super-Sa	31.12	4	0	0.312
11	US-2023-141	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Eneadi	Home Office	United States	Henderson	Kentucky	42420	South	OFF-PA-1000	Office Supply	Paper	Southworth 250; G	6.54	1	0	3.0004
12	US-2023-141	06-01-2023	10-01-2023	Second Class	LS-17230	Lucyann Saunders	Consumer	United States	Los Angeles	California	90043	West	OFF-PA-1000	Office Supply	Paper	Xerox 225	34.44	3	0	9.3312
13	US-2023-141	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Eneadi	Home Office	United States	Henderson	Kentucky	42420	South	TEC-PH-1000	Technology	Phones	Unilever Extender	755.36	4	0	204.109
14	US-2023-141	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Eneadi	Home Office	United States	Henderson	Kentucky	42420	South	TEC-PH-1000	Technology	Phones	GE 30524EE4	391.98	2	0	103.674
15	US-2023-108	07-01-2023	12-01-2023	Standard Class	VS-21620	Wesley Sundaresan	Consumer	United States	Huntsville	Texas	77340	Central	FUR-FU-1000	Furniture	Furnishings	Howard Miller Wa	78.728	3	0.6	-53.71
16	US-2023-108	07-01-2023	12-01-2023	Standard Class	VS-21620	Wesley Sundaresan	Consumer	United States	Huntsville	Texas	77340	Central	OFF-BI-1000	Office Supply	Binders	Acco Four Pocket	10.43	7	0.8	-18.253
17	US-2023-138	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette	Consumer	United States	Laredo	Texas	78041	Central	OFF-AR-1000	Office Supply	Art	Nevel 312	3.344	2	0.2	1.168
18	US-2023-138	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette	Consumer	United States	Laredo	Texas	78041	Central	TEC-AC-1000	Technology	Accessories	Memorex Mono Tra	31.2	3	0.2	9.15
19	US-2023-141	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfield	Virginia	22153	South	FUR-FU-1000	Furniture	Furnishings	Howard Miller T-12	51.94	1	0	21.2564
20	US-2023-141	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfield	Virginia	22153	South	OFF-LA-1000	Office Supply	Labels	Avery 482	2.69	1	0	1.5583
21	US-2023-138	11-01-2023	14-01-2023	First Class	SV-20365	Seeth Vennon	Consumer	United States	Dover	Delaware	19901	East	FUR-FU-1000	Furniture	Furnishings	DAV Value Up-Cha	9.34	2	0	3.0614
22	US-2023-141	11-01-2023	16-01-2023	Standard Class	EO-1805	Brian Dahlen	Consumer	United States	San Francisco	California	94109	West	FUR-BO-1000	Furniture	Bookcases	OT's About Creativ	333.959	3	0.15	3.3594
23	US-2023-109	13-01-2023	16-01-2023	Second Class	NO-9370	Natalie DeCherney	Consumer	United States	Mount Pleasant	South Carolina	29464	South	FUR-CH-1000	Furniture	Chairs	Global Highback L	545.34	6	0	87.5004
24	US-2023-111	13-01-2023	16-01-2023	Standard Class	EO-1805	Brian Dahlen	Consumer	United States	San Francisco	California	94109	West	OFF-AR-1000	Office Supply	Art	4003 Highlighter	13.9	5	0	6.567
25	US-2023-111	13-01-2023	16-01-2023	Standard Class	MM-17920	Michael Moore	Consumer	United States	Newark	Ohio	43005	East	OFF-BI-1000	Office Supply	Binders	Avery HealthPak	3.438	2	0.7	-2.5512
26	US-2023-141	13-01-2023	15-01-2023	Second Class	CS-12250	Chris Selernick	Corporate	United States	Bossier City	Louisiana	70111	South	OFF-AR-1000	Office Supply	Art	3-ring staple pack	5.64	3	0	2.7072
27	US-2023-141	13-01-2023	15-01-2023	Second Class	CS-12250	Chris Selernick	Corporate	United States	Bossier City	Louisiana	70111	South	OFF-BI-1000	Office Supply	Binders	Brown Kraft Respo	50.34	3	0	25.47
28	US-2023-141	13-01-2023	15-01-2023	Second Class	CS-12250	Chris Selernick	Corporate	United States	Bossier City	Louisiana	70111	South	OFF-EN-1000	Office Supply	Envelopes	Staple envelope	11.36	2	0	5.3302
29	US-2023-111	13-01-2023	16-01-2023	Standard Class	MM-17920	Michael Moore	Consumer	United States	Newark	Ohio	43005	East	OFF-PA-1000	Office Supply	Paper	Xerox 9223	37.408	7	0.2	10.0028
30	US-2023-141	13-01-2023	15-01-2023	Second Class	CS-12250	Chris Selernick	Corporate	United States	Bossier City	Louisiana	70111	South	OFF-ST-1000	Office Supply	Storage	Fellowes StoDraw	572.58	6	0	34.5486
31	US-2023-141	13-01-2023	16-01-2023	Standard Class	EO-1805	Brian Dahlen	Consumer	United States	San Francisco	California	94109	West	OFF-ST-1000	Office Supply	Storage	Tennisco 6-in B	1325.95	5	0	238.853
32	US-2023-141	13-01-2023	15-01-2023	Second Class	CS-12250	Chris Selernick	Corporate	United States	Bossier City	Louisiana	70111	South	TEC-AC-1000	Technology	Accessories	Planonics STC Co	646.74	6	0	258.868
33	US-2023-141	14-01-2023	15-01-2023	First Class	BS-15580	Brendan Sreed	Corporate	United States	Philadelphia	Pennsylvania	19104	East	FUR-BO-1000	Furniture	Bookcases	Sauder Cornercon	61.96	4	0.5	-53.288
34	US-2023-109	15-01-2023	17-01-2023	First Class	EH-19550	Erica Hachney	Consumer	United States	Roswell	Georgia	30076	South	TEC-AC-1000	Technology	Accessories	Logitech 910-002	148.85	5	0	65.570
35	US-2023-111	16-01-2023	18-01-2023	Second Class	DL-13315	Dellina Luchford	Consumer	United States	Philadelphia	Pennsylvania	19104	East	FUR-FU-1000	Furniture	Furnishings	DAV Black Cherry	127.104	6	0.2	28.5584
36	US-2023-111	16-01-2023	18-01-2023	Second Class	DL-13315	Dellina Luchford	Consumer	United States	Philadelphia	Pennsylvania	19104	East	OFF-BI-1000	Office Supply	Binders	GBG LivenBinding	18.588	2	0.7	-13.631
37	US-2023-111	16-01-2023	18-01-2023	Second Class	DL-13315	Dellina Luchford	Consumer	United States	Philadelphia	Pennsylvania	19104	East	OFF-LA-1000	Office Supply	Labels	Round Specialty L	30.072	3	0.2	10.8433
38	US-2023-111	16-01-2023	18-01-2023	Second Class	DL-13315	Dellina Luchford	Consumer	United States	Philadelphia	Pennsylvania	19104	East	TEC-PH-1000	Technology	Phones	AT&T 184000 Phoe	94.2	3	0.4	-31.05
39	US-2023-132	18-01-2023	21-01-2023	Second Class	DV-1195	David Warner	Corporate	United States	Springfield	Oregon	97477	West	OFF-AP-1000	Office Supply	Appliances	Fellowes Mglay B	64.864	4	0.2	6.4864
40	US-2023-141	19-01-2023	20-01-2023	First Class	TS-21340	Toby Swindell	Consumer	United States	Scottsdale	Arizona	85254	West	FUR-BO-1000	Furniture	Bookcases	OT's Urban 4-Shel	1814.7	5	0.7	-320.6
41	US-2023-141	19-01-2023	20-01-2023	First Class	TS-21340	Toby Swindell	Consumer	United States	Scottsdale	Arizona	85254	West	OFF-BI-1000	Office Supply	Binders	GBG StandardPlan	32.34	10	0.7	-323.786
42	US-2023-141	19-01-2023	20-01-2023	First Class	TS-21340	Toby Swindell	Consumer	United States	Scottsdale	Arizona	85254	West	OFF-EN-1000	Office Supply	Envelopes	Tywel Top-Oper	108.72	5	0.2	36.693
43	US-2023-141	19-01-2023	20-01-2023	First Class	TS-21340	Toby Swindell	Consumer	United States	Scottsdale	Arizona	85254	West	OFF-PA-1000	Office Supply	Paper	TOPS Carbonless	56.064	4	0.2	19.6224
44	US-2023-141	20-01-2023	26-01-2023	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Jonesboro	Arkansas	72401	South	FUR-CH-1000	Furniture	Chairs	Hon 4700 Series	1007.34	3	0	224.267
45	US-2023-141	20-01-2023	26-01-2023	Standard Class	XP-21885	Xylona Prais	Consumer	United States	Westland	Michigan	48185	Central	FUR-FU-1000	Furniture	Furnishings	Ultra Dosh Push P	14.73	3	0	4.8609
46	US-2023-141	20-01-2023	26-01-2023	Standard Class	XP-21885	Xylona Prais	Consumer	United States	Westland	Michigan	48185	Central	FUR-FU-1000	Furniture	Furnishings	Edson ClustersMa	272.34	3	0	30.0234
47	US-2023-141	20-01-2023	26-01-2023	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Jonesboro	Arkansas	72401	South	FUR-FU-1000	Furniture	Furnishings	Edson Expressions	138.6	4	0	11.95
48	US-2023-141	20-01-2023	26-01-2023	Standard Class	MV-17485	Mark Van Hult	Consumer	United States	Los Angeles	California	90043	West	FUR-FU-1000	Furniture	Furnishings	Edson Expressions	113.3	2	0	5.79
49	US-2023-141	20-01-2023	26-01-2023	Standard Class	XP-21885	Xylona Prais	Consumer	United States	Westland	Michigan	48185	Central	OFF-AP-1000	Office Supply	Appliances	Holmes Replaceme	247.76	4	0.1	93.5896
50	US-2023-141	20-01-2023	26-01-2023	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Jonesboro	Arkansas	72401	South	OFF-AR-1000	Office Supply	Art	Nevel 351	22.36	7	0	6.8564
51	US-2023-141	20-01-2023	26-01-2023	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Jonesboro	Arkansas	72401	South	OFF-AR-1000	Office Supply	Art	Nevel 327	6.63	3	0	1.7901
52	US-2023-141	20-01-2023	26-01-2023	Standard Class	XP-21885	Xylona Prais	Consumer	United States	Westland	Michigan	48185	Central	OFF-AR-1000	Office Supply	Art	Panasonic KP-4A	43.32	3	0	12.768

Experiment 1 — Text Table

Aim: To display category-wise sales figures in a simple tabular format.

Procedure:

- Drag Category to Rows.
- Drag Sales to Text on the Marks card.
- From Show Me, select Text Table.

Configuration / Field Placement: Category → Rows; Sales → Text.

Expected Output: A grid listing each product category alongside its total sales value.

Business Insight: Compare category-wise sales.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 1 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 2 — Heat Map

Aim: To visualise sales intensity across regions and categories using colour.

Procedure:

- Drag Region to Columns.
- Drag Category to Rows.
- Drag Sales to Color.
- Select Heat Map from Show Me.

Configuration / Field Placement: Region → Columns; Category → Rows; Sales → Color.

Expected Output: A colour-graded grid where darker/brighter cells indicate higher sales.

Business Insight: Identify high-performing regions.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 2 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 3 — Highlight Table

Aim: To highlight profitable and loss-making category/region combinations.

Procedure:

- Drag Region to Columns.
- Drag Category to Rows.
- Drag Profit to Color.
- Select Highlight Table from Show Me.

Configuration / Field Placement: Region → Columns; Category → Rows; Profit → Color.

Expected Output: A shaded table showing exact profit figures with colour-coded performance.

Business Insight: Locate profitable and loss-making segments.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 3 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 4 — Symbol Map

Aim: To plot the geographic distribution of sales and profit by state.

Procedure:

- Drag State to Detail.
- Drag Sales to Size.
- Drag Profit to Color.
- Select Symbol Map from Show Me.

Configuration / Field Placement: State → Detail; Sales → Size; Profit → Color.

Expected Output: A US map with circles sized by sales and coloured by profit for each state.

Business Insight: Reveal geographical sales distribution.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 4 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 5 — Bar Chart

Aim: To compare total sales across product categories.

Procedure:

- Drag Category to Rows.
- Drag Sales to Columns.
- Select Bar Chart from Show Me.

Configuration / Field Placement: Category → Rows; Sales → Columns.

Expected Output: Horizontal bars, one per category, ranked by total sales.

Business Insight: Compare category performance.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 5 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 6 — Filled Map

Aim: To show state-wise sales penetration using a shaded map.

Procedure:

- Drag State to Detail.
- Drag Sales to Color.
- Select Filled Map from Show Me.

Configuration / Field Placement: State → Detail; Sales → Color.

Expected Output: A US map with each state shaded by total sales value.

Business Insight: Show state-wise market penetration.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 6 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 7 — Stacked Bar Chart

Aim: To show how each customer segment contributes to category sales.

Procedure:

- Drag Category to Rows.
- Drag Sales to Columns.
- Drag Segment to Color.
- Select Stacked Bars from Show Me.

Configuration / Field Placement: Category → Rows; Sales → Columns; Segment → Color.

Expected Output: Bars per category, each divided into coloured segments for Consumer, Corporate and Home Office.

Business Insight: Show segment contribution to sales.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 7 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 8 — Side-by-Side Bar Chart

Aim: To compare segment sales within each category directly, rather than stacked.

Procedure:

- Drag Category to Rows.
- Drag Sales to Columns.
- Drag Segment to Columns.
- Select Side-by-Side Bars from Show Me.

Configuration / Field Placement: Category → Rows; Sales → Columns; Segment → Columns.

Expected Output: Grouped bars, one cluster per category, with a separate bar per segment.

Business Insight: Enable segment comparison.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 8 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 9 — Line Chart

Aim: To track the sales trend over the period covered by the dataset.

Procedure:

- Drag Order Date to Columns.
- Drag Sales to Rows.
- Select Line Chart from Show Me.

Configuration / Field Placement: Order Date → Columns; Sales → Rows.

Expected Output: A continuous line showing sales rising and falling across the date range.

Business Insight: Support sales trend analysis.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 9 on Dashboard 1, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 10 — Dual Line Chart

Aim: To compare the sales trend and the profit trend on one chart.

Procedure:

- Drag Order Date to Columns.
- Drag Sales to Rows.
- Drag Profit to Rows.
- Set the second measure as a Dual Axis.

Configuration / Field Placement: Order Date → Columns; Sales → Rows; Profit → Rows (Dual Axis).

Expected Output: Two overlaid lines, one for sales and one for profit, sharing the same date axis.

Business Insight: Show the relationship between sales and profit.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 10 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 11 — Dual Line Chart (Extended View)

Aim: To re-examine the sales/profit relationship at a finer time granularity for deeper analysis.

Procedure:

- Drag Order Date to Columns.
- Drag Sales to Rows.

- Drag Profit to Rows.
- Set the second measure as a Dual Axis.

Configuration / Field Placement: Order Date → Columns; Sales → Rows; Profit → Rows (Dual Axis).

Expected Output: A second dual-axis view, used on Dashboard 2 to support the profitability analysis carried over from Sheet 10.

Business Insight: Reinforce the relationship between sales and profit at a granular level.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 11 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 12 — Pie Chart

Aim: To show each category's share of total sales.

Procedure:

- Drag Category to Color.
- Drag Sales to Angle.
- Select Pie Chart from Show Me.

Configuration / Field Placement: Category → Color; Sales → Angle.

Expected Output: A pie divided into wedges, one per category, sized by its share of total sales.

Business Insight: Show category contribution to overall sales.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 12 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 13 — Scatter Plot

Aim: To examine the relationship between sales and profit at the category level.

Procedure:

- Drag Sales to Columns.
- Drag Profit to Rows.
- Drag Category to Color.
- Select Scatter Plot from Show Me.

Configuration / Field Placement: Sales → Columns; Profit → Rows; Category → Color.

Expected Output: Individual marks plotted by sales (x-axis) against profit (y-axis), coloured by category.

Business Insight: Support profitability analysis and outlier detection.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 13 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 14 — Circle View

Aim: To compare sales and profit across categories in one compact visual.

Procedure:

- Drag Category to Rows.
- Drag Sales to Size.
- Drag Profit to Color.
- Select Circle View from Show Me.

Configuration / Field Placement: Category → Rows; Sales → Size; Profit → Color.

Expected Output: A row of circles per category, sized by sales and coloured by profit.

Business Insight: Provide a compact visual comparison of two measures.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 14 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 15 — Histogram

Aim: To study the distribution of profit across all order lines.

Procedure:

- Select the Profit field.
- Select Histogram from Show Me.

Configuration / Field Placement: Profit → Select (binned automatically).

Expected Output: A bar chart of profit bins on the x-axis against the count of order lines on the y-axis.

Business Insight: Reveal the distribution of profit, including the frequency of losses.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 15 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 16 — Gantt Chart

Aim: To compare individual orders by their sales value along a sequential view.

Procedure:

- Drag Order ID to Rows.
- Drag Sales to Size.
- Select Gantt Chart from Show Me.

Configuration / Field Placement: Order ID → Rows; Sales → Size.

Expected Output: A row per order, with bar length/size reflecting the order's sales value.

Business Insight: Support order-level comparison.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 16 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 17 — Filters

Aim: To make the dashboard's worksheets interactively filterable by region.

Procedure:

- Drag Region to the Filters shelf.
- Select the regions to include.
- Right-click the filter and choose Show Filter.

Configuration / Field Placement: Region → Filters (with the filter card displayed on screen).

Expected Output: An on-screen Region filter control that a viewer can use to narrow every connected worksheet.

Business Insight: Enable interactive dashboard filtering.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 17 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

Experiment 18 — Creating Business KPIs in Tableau — KPI: Total Sales

Aim: To create a headline KPI card displaying total sales for the dashboard.

Procedure:

- Open a new worksheet and rename it “KPI - Total Sales”.
- Drag Sales from Measures to the Text mark card.
- Tableau displays the aggregated total sales value.
- Click the Sales field on the Marks card → Format.
- Format the number as Currency (Custom).
- Increase the font size (20–30 pt) and make it bold.

Configuration / Field Placement: Sales → Text (Marks card), formatted as bold Currency at 20–30 pt.

Expected Output: A single, large, currency-formatted number showing total sales.

Business Insight: Give the dashboard a clear, unmissable headline metric.

Result: The worksheet was successfully created in Tableau Desktop and included as Sheet 18 on Dashboard 2, in line with the practical brief grouping Sheets 1–9 into the first dashboard and Sheets 10–18 into the second.

TABLEAU DASHBOARD PUBLISHING AND ANALYSIS

This section documents the publishing and analysis stage of the two practical dashboards built from Experiments 1–18, alongside the two independently developed dashboard projects referenced earlier in this report.

Dashboard 1 — Sales Intelligence Dashboard (Sheets 1–9)

Dashboard screenshot, reproduced below as Figure D.1.

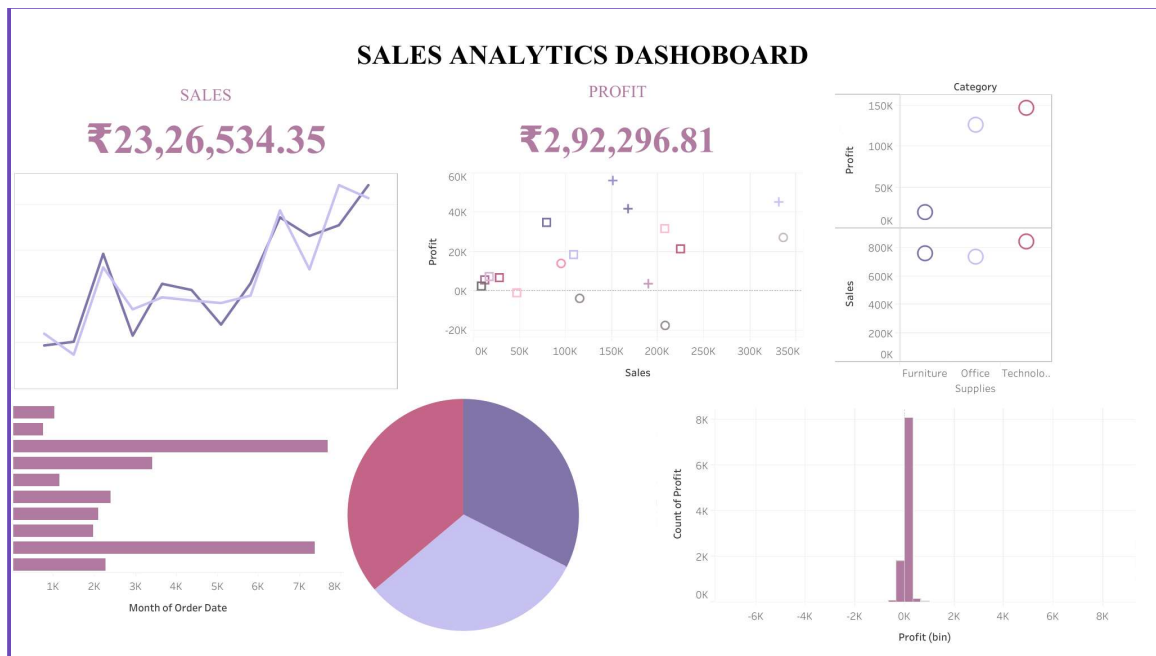


Figure D.1 — Sales Intelligence / Sales Analytics Dashboard (Tableau)

Dashboard title: Sales Intelligence Dashboard.

Purpose: To bring together the foundational chart types (Text Table, Heat Map, Highlight Table, Symbol Map, Bar Chart, Filled Map, Stacked Bar Chart, Side-by-Side Bar Chart and Line Chart) into one interactive view of category, regional and time-based sales performance.

KPIs: Total Sales (via the KPI card from Experiment 18, if placed on this dashboard) alongside the category and regional sales comparisons from Sheets 1–9.

Charts used: Sheets 1 to 9, as documented in the Practical Programs section.

Filters: Region filter (Experiment 17), where included on this dashboard.

Business insights: to be inserted by student, once the published workbook is reviewed against the specific figures it displays.

Management interpretation: to be inserted by student.

Dashboard 2 — Product Sales Analysis Dashboard (Sheets 10–18)

Dashboard screenshot, reproduced below as Figure D.2.

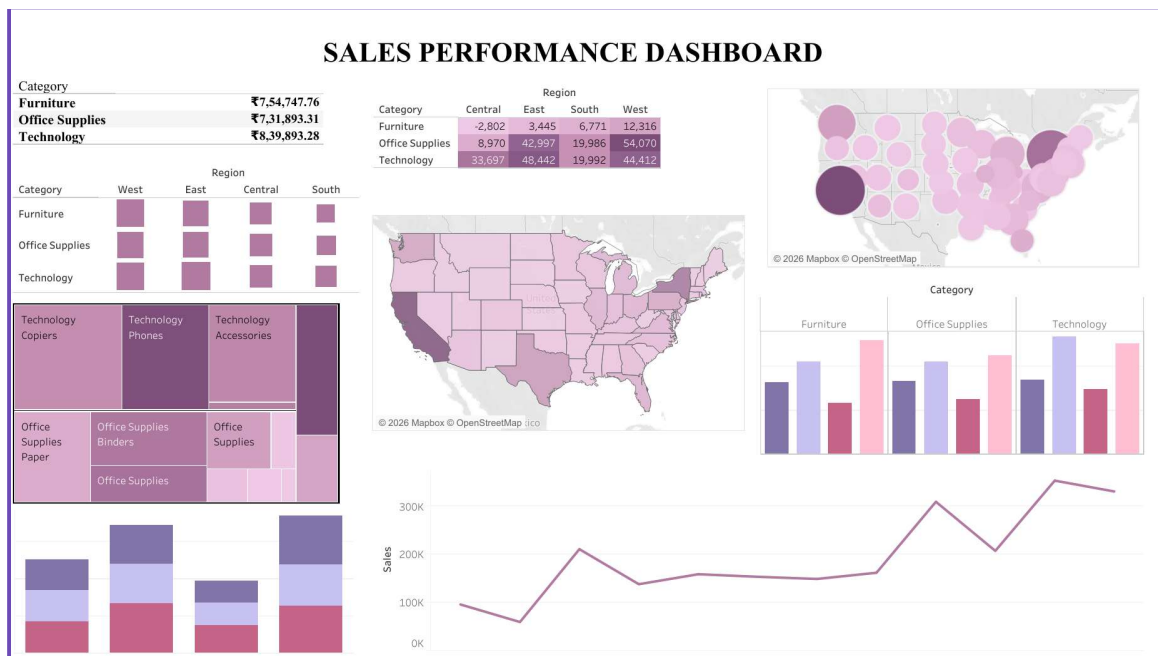


Figure D.2 — Product Sales Analysis / Sales Performance Dashboard (Tableau)

Dashboard title: Product Sales Analysis Dashboard.

Purpose: To extend the analysis into profitability, distribution and order-level detail using the Dual Line Chart, Pie Chart, Scatter Plot, Circle View, Histogram, Gantt Chart, Filters and the Total Sales KPI card (Sheets 10–18).

KPIs: Total Sales (via the KPI card from Experiment 18, if placed on this dashboard).

Charts used: Sheets 10 to 18, as documented in the Practical Programs section.

Filters: Region filter (Experiment 17), where included on this dashboard.

Business insights: to be inserted by student.

Management interpretation: to be inserted by student.

DASHBOARD PRINTOUTS

This section is reserved for the actual dashboard screenshots and printouts taken from the two published Tableau Public dashboards described in this report.

To be inserted by student:

- Full-page printout of the Sales Intelligence Dashboard (Sheets 1–9), exported from Tableau Public.
- Full-page printout of the Product Sales Analysis Dashboard (Sheets 10–18), exported from Tableau Public.
- Optional: individual worksheet screenshots for any chart the student wishes to highlight separately.

Two dashboard screenshots, sourced from the project repository, are reproduced in full below as the dashboard printouts. Individual worksheet-level exports were not supplied and are left for the student to add separately if desired.

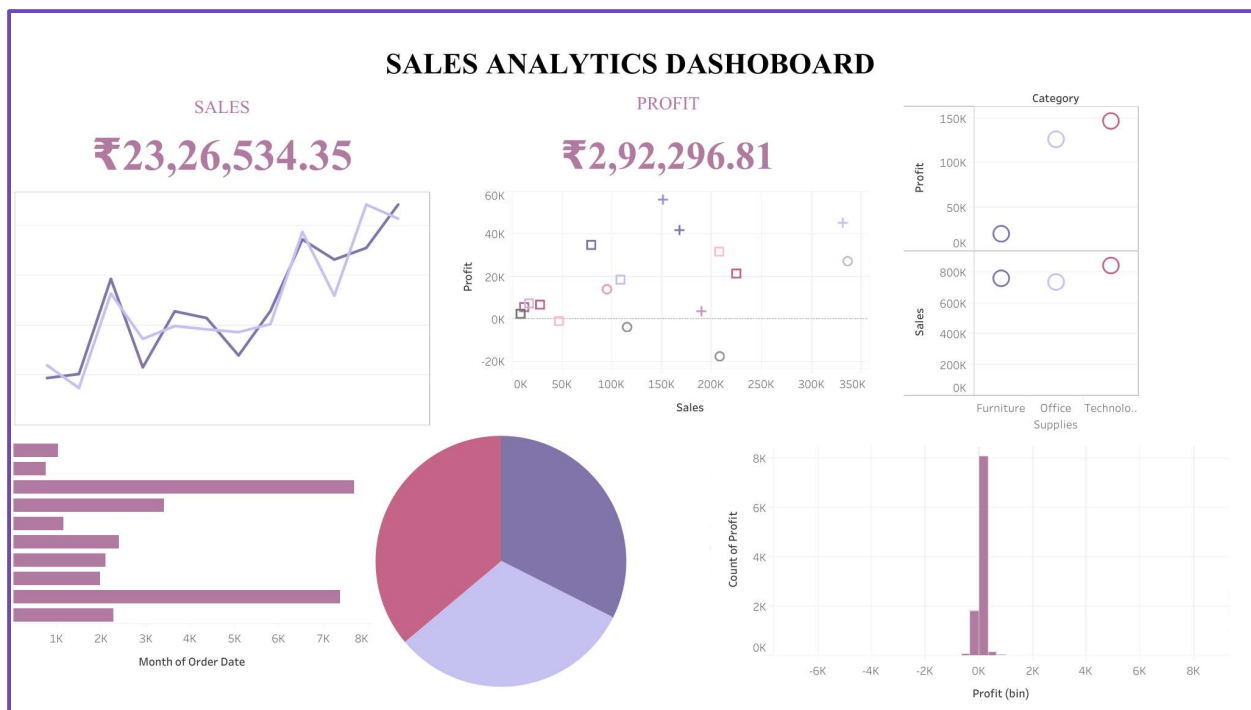


Figure P.1 — Full-Page Printout: Sales Intelligence / Sales Analytics Dashboard

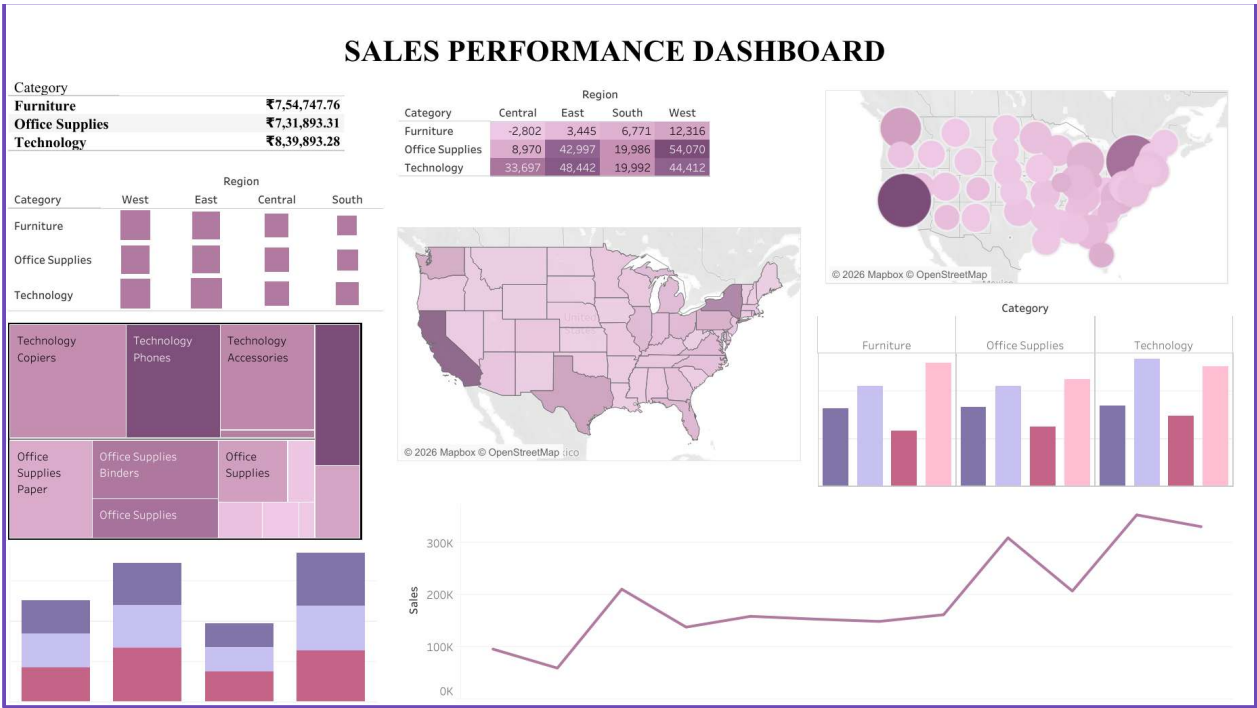


Figure P.2 — Full-Page Printout: Product Sales Analysis / Sales Performance Dashboard

DASHBOARD LINKS

Dashboard	Platform	Link
Sales Intelligence Dashboard	Tableau Public	https://public.tableau.com/app/profile/rakshitha.m.m4396/viz/SalesAnalyticsDashboard_17857620962110/SalesAnalytics
Product Sales Dashboard	Tableau Public	https://public.tableau.com/app/profile/rakshitha.m.m4396/viz/SalesPerformanceDashboard_17857583223040/Dashboard1

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- Tableau Public. Public gallery and publishing platform documentation, tableau.com/public.